

Newman-Penrose Formalism Calculations

Stephani et al., Equation (28.45)

Metric

$$ds^2 = \frac{r^2}{P_2(m, r)^2} du \otimes du - du \otimes dr - r^2 du \otimes dm - dr \otimes du - r^2 dm \otimes du + r^2 P(m)^2 dm \otimes dm + \frac{r^2}{P(m)^2} dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, r, m, y)$$

Metric Functions

$$P^{-2} = -\frac{1}{2}m^4 + am^2 + bm + c$$

$$P_2 = -\frac{1}{2}m^4 + am^2 + bm + \frac{2m^3}{r} + c - \frac{2am}{r} - \frac{3m^2}{r^2} - \frac{b}{r} + \frac{a}{r^2} + \frac{2m}{r^3} - \frac{1}{2r^4}$$

Null Tetrad

Covariant components

$$l = (-1) du$$

$$n = \left(\frac{r^2}{2P_2(m, r)^2} \right) du + (-1) dr + (-r^2) dm$$

$$m = \left(\frac{1}{2} \sqrt{2} r P(m) \right) dm + \left(\frac{i \sqrt{2} r}{2 P(m)} \right) dy$$

$$\overline{m} = \left(\frac{1}{2} \sqrt{2} r P(m) \right) dm + \left(-\frac{i \sqrt{2} r}{2 P(m)} \right) dy$$

Contravariant components

$$l = (1) dr$$

$$n = (1) du + \left(\frac{r^2}{2 P_2(m, r)^2} \right) dr$$

$$m = \left(-\frac{\sqrt{2} r}{2 P(m)} \right) dr + \left(\frac{\sqrt{2}}{2 r P(m)} \right) dm + \left(\frac{i \sqrt{2} P(m)}{2 r} \right) dy$$

$$\overline{m} = \left(-\frac{\sqrt{2} r}{2 P(m)} \right) dr + \left(\frac{\sqrt{2}}{2 r P(m)} \right) dm + \left(-\frac{i \sqrt{2} P(m)}{2 r} \right) dy$$

Directional Derivatives

$$D = \partial_r$$

$$\Delta = \partial_u + \frac{r^2}{2 P_2(m, r)^2} \partial_r$$

$$\delta = -\frac{\sqrt{2} r}{2 P(m)} \partial_r + \frac{\sqrt{2}}{2 r P(m)} \partial_m + \frac{i \sqrt{2} P(m)}{2 r} \partial_y$$

$$\overline{\delta} = -\frac{\sqrt{2} r}{2 P(m)} \partial_r + \frac{\sqrt{2}}{2 r P(m)} \partial_m - \frac{i \sqrt{2} P(m)}{2 r} \partial_y$$

Spin Coefficients

$$\rho = -\frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{r}{2\,P_2\left(m,r\right)^2}$$

$$\lambda = 0$$

$$\nu = -\frac{\sqrt{2}r^3\frac{\partial}{\partial r}P_2\left(m,r\right)}{2\,P\left(m\right)P_2\left(m,r\right)^3} + \frac{\sqrt{2}r\frac{\partial}{\partial m}P_2\left(m,r\right)}{2\,P\left(m\right)P_2\left(m,r\right)^3}$$

$$\alpha = \frac{\sqrt{2}}{2\,P\left(m\right)} + \frac{\sqrt{2}\frac{\partial}{\partial m}P\left(m\right)}{4\,rP\left(m\right)^2}$$

$$\beta = -\frac{\sqrt{2}\frac{\partial}{\partial m}P\left(m\right)}{4\,rP\left(m\right)^2}$$

$$\tau = \frac{\sqrt{2}}{2\,P\left(m\right)}$$

$$\pi = -\frac{\sqrt{2}}{2\,P\left(m\right)}$$

$$\epsilon = 0$$

$$\gamma = \frac{r^2 \frac{\partial}{\partial r} P_2(m, r)}{2 P_2(m, r)^3} - \frac{r}{2 P_2(m, r)^2}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{3r^2 \frac{\partial}{\partial r} P_2(m, r)^2}{4 P_2(m, r)^4} + \frac{r^2 \frac{\partial^2}{(\partial r)^2} P_2(m, r)}{4 P_2(m, r)^3} + \frac{r \frac{\partial}{\partial r} P_2(m, r)}{P_2(m, r)^3} - \frac{\frac{\partial}{\partial m} P(m)}{2r P(m)^3} - \frac{3 \frac{\partial}{\partial m} P(m)^2}{4r^2 P(m)^4} + \frac{\frac{\partial^2}{(\partial m)^2} P(m)}{4r^2 P(m)^3}$$

$$\Phi_{12} = \frac{3\sqrt{2}r^3 \frac{\partial}{\partial r} P_2(m, r)^2}{4 P(m) P_2(m, r)^4} - \frac{\sqrt{2}r^3 \frac{\partial^2}{(\partial r)^2} P_2(m, r)}{4 P(m) P_2(m, r)^3} - \frac{\sqrt{2}r^2 \frac{\partial}{\partial r} P_2(m, r)}{P(m) P_2(m, r)^3} + \frac{\sqrt{2}r \frac{\partial^2}{\partial m \partial r} P_2(m, r)}{4 P(m) P_2(m, r)^3} - \frac{3\sqrt{2}r \frac{\partial}{\partial m} P_2(m, r) \frac{\partial}{\partial r} P_2(m, r)}{4 P(m) P_2(m, r)^4} + \frac{\sqrt{2} \frac{\partial}{\partial m} P_2(m, r)}{2 P(m) P_2(m, r)^3}$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = \frac{3\sqrt{2}r^3 \frac{\partial}{\partial r} P_2(m, r)^2}{4 P(m) P_2(m, r)^4} - \frac{\sqrt{2}r^3 \frac{\partial^2}{(\partial r)^2} P_2(m, r)}{4 P(m) P_2(m, r)^3} - \frac{\sqrt{2}r^2 \frac{\partial}{\partial r} P_2(m, r)}{P(m) P_2(m, r)^3} + \frac{\sqrt{2}r \frac{\partial^2}{\partial m \partial r} P_2(m, r)}{4 P(m) P_2(m, r)^3} - \frac{3\sqrt{2}r \frac{\partial}{\partial m} P_2(m, r) \frac{\partial}{\partial r} P_2(m, r)}{4 P(m) P_2(m, r)^4} + \frac{\sqrt{2} \frac{\partial}{\partial m} P_2(m, r)}{2 P(m) P_2(m, r)^3}$$

$$\begin{aligned} \Phi_{22} = & -\frac{3r^4 \frac{\partial}{\partial r} P_2(m, r)^2}{2 P(m)^2 P_2(m, r)^4} + \frac{r^4 \frac{\partial^2}{(\partial r)^2} P_2(m, r)}{2 P(m)^2 P_2(m, r)^3} + \frac{2r^3 \frac{\partial}{\partial r} P_2(m, r)}{P(m)^2 P_2(m, r)^3} - \frac{r^2 \frac{\partial^2}{\partial m \partial r} P_2(m, r)}{P(m)^2 P_2(m, r)^3} + \frac{r^2 \frac{\partial}{\partial m} P(m) \frac{\partial}{\partial r} P_2(m, r)}{P(m)^3 P_2(m, r)^3} \\ & + \frac{3r^2 \frac{\partial}{\partial m} P_2(m, r) \frac{\partial}{\partial r} P_2(m, r)}{P(m)^2 P_2(m, r)^4} - \frac{r \frac{\partial}{\partial m} P_2(m, r)}{P(m)^2 P_2(m, r)^3} - \frac{\frac{\partial}{\partial m} P(m) \frac{\partial}{\partial m} P_2(m, r)}{P(m)^3 P_2(m, r)^3} - \frac{3 \frac{\partial}{\partial m} P_2(m, r)^2}{2 P(m)^2 P_2(m, r)^4} + \frac{\frac{\partial^2}{(\partial m)^2} P_2(m, r)}{2 P(m)^2 P_2(m, r)^3} \end{aligned}$$

$$\Lambda = \frac{r^2 \frac{\partial}{\partial r} P_2(m, r)^2}{4 P_2(m, r)^4} - \frac{r^2 \frac{\partial^2}{(\partial r)^2} P_2(m, r)}{12 P_2(m, r)^3} - \frac{2r \frac{\partial}{\partial r} P_2(m, r)}{3 P_2(m, r)^3} - \frac{1}{2 P(m)^2} + \frac{1}{2 P_2(m, r)^2} - \frac{\frac{\partial}{\partial m} P(m)}{2r P(m)^3} - \frac{\frac{\partial}{\partial m} P(m)^2}{4r^2 P(m)^4} + \frac{\frac{\partial^2}{(\partial m)^2} P(m)}{12r^2 P(m)^3}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{r^2 \frac{\partial}{\partial r} P_2(m, r)^2}{2 P_2(m, r)^4} + \frac{r^2 \frac{\partial^2}{(\partial r)^2} P_2(m, r)}{6 P_2(m, r)^3} + \frac{r \frac{\partial}{\partial r} P_2(m, r)}{3 P_2(m, r)^3} + \frac{\frac{\partial}{\partial m} P(m)^2}{2r^2 P(m)^4} - \frac{\frac{\partial^2}{(\partial m)^2} P(m)}{6r^2 P(m)^3}$$

$$\Psi_3 = \frac{3\sqrt{2}r^3 \frac{\partial}{\partial r} P_2(m, r)^2}{4 P(m) P_2(m, r)^4} - \frac{\sqrt{2}r^3 \frac{\partial^2}{(\partial r)^2} P_2(m, r)}{4 P(m) P_2(m, r)^3} - \frac{\sqrt{2}r^2 \frac{\partial}{\partial r} P_2(m, r)}{2 P(m) P_2(m, r)^3} + \frac{\sqrt{2}r \frac{\partial^2}{\partial m \partial r} P_2(m, r)}{4 P(m) P_2(m, r)^3} - \frac{3\sqrt{2}r \frac{\partial}{\partial m} P_2(m, r) \frac{\partial}{\partial r} P_2(m, r)}{4 P(m) P_2(m, r)^4}$$

$$\Psi_4 = -\frac{3r^4 \frac{\partial}{\partial r} P_2(m, r)^2}{2 P(m)^2 P_2(m, r)^4} + \frac{r^4 \frac{\partial^2}{(\partial r)^2} P_2(m, r)}{2 P(m)^2 P_2(m, r)^3} + \frac{r^3 \frac{\partial}{\partial r} P_2(m, r)}{P(m)^2 P_2(m, r)^3} - \frac{r^2 \frac{\partial^2}{\partial m \partial r} P_2(m, r)}{P(m)^2 P_2(m, r)^3} + \frac{3r^2 \frac{\partial}{\partial m} P_2(m, r) \frac{\partial}{\partial r} P_2(m, r)}{P(m)^2 P_2(m, r)^4} - \frac{3 \frac{\partial}{\partial m} P_2(m, r)^2}{2 P(m)^2 P_2(m, r)^4} + \frac{\frac{\partial^2}{(\partial m)^2} P_2(m, r)}{2 P(m)^2 P_2(m, r)^3}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"